**12** Which of the following statements is **false** in describing a mass moving in simple harmonic motion?

**A** The maximum kinetic energy is dependent on the frequency of the oscillation.

**B** The time taken for the system to change from maximum kinetic energy to maximum potential energy is a quarter of the period of the oscillation.

**C** An oscillating system with larger amplitude will have a longer period.

**D** An oscillating system with larger amplitude will have a greater maximum velocity.